Shriram Raja

Boston, MA

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RESEARCH INTERESTS

Scheduling, Heterogeneous Architectures, Resource Management, Synchronization, Virtualization

EDUCATION

Boston University, Boston, MA — Sep 2023 - Present
Ph.D. in Computer Science

- Advisor: Dr. Richard West

Virginia Tech, Blacksburg, VA — Aug 2021 - May 2023
Master of Engineering, Computer Engineering

- Advisor: Dr. Haibo Zeng
- Project Title: Hybrid Priority Assignment for Global Fixed Priority Scheduling

PSG College of Technology, Coimbatore, India — Aug 2017 - May 2021
Bachelor of Engineering, Electrical & Electronics Engineering

PUBLICATIONS

[P1] **Shriram Raja***, Zhiyuan Ruan*, and Richard West, "Pomegranate: A Lightweight Compartmentalization Architecture using Virtualization Extensions", *arxiv*, 2026 [arxiv]

[A1] Richard West, Zhiyuan Ruan, **Shriram Raja**, and Rafiuddin Syed, "Tutorial: Mixed-Criticality Computing with the Quest RTOS and Quest-V Partitioning Hypervisor", *25th ACM SIGBED International Conference on Embedded Software (EMSOFT)*, 2025 [Paper][DOI]

[J1] Xuanliang Deng*, **Shriram Raja***, Yecheng Zhao, and Haibo Zeng, "Priority Assignment for Global Fixed Priority Scheduling on Multiprocessors", *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2024 [Paper][Code][DOI]

* - contributed equally

EXPERIENCE

Boston University — Sep 2023 - Present
Research Fellow — *Boston, MA*

- Developed bounded priority-aware kernel locks for Quest RTOS
- Compartmentalized the Linux kernel using hardware virtualization features on x86 processors as a part of the DARPA CPM program with fellow Ph.D. student Zhiyuan (Ryan) Ruan
- Contributed to the Quest-V Software Development Kit which supports Quest and Yocto Linux guests

Virginia Tech — Mar 2022 - May 2023
Research Assistant — *Blacksburg, VA*

- Developed, implemented and evaluated a novel Global Fixed Priority scheduling algorithm for multicore real-time systems, advised by Dr. Haibo Zeng leading to the publication in IEEE TCAD 2024

Security Solutions, Marvell Semiconductor — May - Aug 2022
Firmware Engineer Intern — *Santa Clara, CA*

- Implemented FRAM Logging feature in LiquidSecurity Cloud Hardware Security Module to collect debug information during boot-up

TEACHING EXPERIENCE

CS 553 Advanced Operating Systems, Guest Lecturer, Boston University — Spring 2026
CS 410 Advanced Software Systems, Teaching Fellow, Boston University — Spring 2025
CS 552 Operating Systems, Teaching Fellow, Boston University — Fall 2024
ECE 5480 Cybersecurity & IoT, Grad Teaching Assistant, Virginia Tech — Spring 2022, 2023, Fall 2022

SERVICE AND PRESENTATIONS

Mentoring

- Varsha Rajkumar: MS CS @ BU Spring 2026 (Next: Ph.D. Student, MPI-SWS)
  - **Topic:** Fault Detection in the Quest-V Hypervisor

Reviewing

- **Secondary Reviewer**
  - Software: Practice and Experience — 2025
  - IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS) — 2025
  - IEEE Real-Time Systems Symposium (RTSS) — 2024
- **Artifact Evaluation**
  - USENIX Symposium on Operating Systems Design and Implementation (OSDI) — 2026
  - IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS) (Secondary) 2026
- **Shadow Technical Program Committee**
  - Euromicro Conference on Real-Time Systems (ECRTS) — 2025, 2026

Tutorial

- Getting Started with the Quest RTOS & Quest-V Hypervisor RTSS 2024, ESWEEK 2025, RTSS 2025

ACADEMIC PROJECTS

i386 Custom Bare-metal Operating System — Fall 2023

Developed a bare-metal operating system for 32-bit x86 systems that uses preemptive FIFO scheduling and supports a simple file system.

Scheduler for Real-Time Operating Systems — Apr 2022

Programmed different periodic scheduling algorithms, resource management protocols, and a polling server for FreeRTOS. Determined the best priority that can be assigned to the polling server by analyzing the response time of aperiodic tasks by varying periodic load for different test cases when executed on an Arduino Mega 2560.

LLVM Optimization Pass — Apr 2022

Implemented the Lazy Code Motion (LCM) algorithm to eliminate redundant statements and move arithmetic expressions to the latest point in the program without modifying the functionality of the code. Evaluated the performance of the optimization pass using open-source benchmarks.

CPU Profiling Tool for Linux — Nov 2021

Used Kprobe to track the cumulative run time and the number of times each task is scheduled by the Linux scheduler. Created and maintained a red-black tree that stores the run time of the tasks. The 20 most scheduled tasks are identified, and their stack trace is displayed using the /proc file system.